CODEFUSION: PYTHON & JAVA SHOWDOWN
PYTHON & JAVA-BASED CODING COMPETITION
ROUND 1 - DEBUGGING CHALLENGE

Team Name		**Department**	
Student Name 1		**Student Name 2**	
Class & Year		**Date**	**22/03/2025 (Saturday)**
Language Chosen (Circle One)s		○ **Python**	○ **Java**

Instructions:

- Identify and fix the errors in the given code.
- Write the corrected version below each question.
- Do not change logic unless necessary for correctness.

Question 1: Sum of Digits of a Number

Buggy Code (Python)	Buggy Code (Java)
def sum_of_digits(n): sum = 0 while n > 0: sum += n % 10 n // 10 return sum print(sum_of_digits(123)) **# Expected Output: 6**	class Main { public static int sumOfDigits(int n) { int sum = 0; while (n > 0) { sum += n % 10; n / 10; } return sum; } public static void main(String[] args) { System.out.println(sumOfDigits(123)); **// Expected Output: 6** } }
Corrected Code	**Output**

Question 2: Reverse a String

Buggy Code (Python)	Buggy Code (Java)
def reverse_string(s): return s[::-2] print(reverse_string("hello")) **# Expected Output: "olleh"**	class Main { public static String reverseString(String s) { return new StringBuilder(s).reverse().toString; } public static void main(String[] args) { System.out.println(reverseString("hello")); } } **// Expected Output: "olleh"**
Corrected Code	**Output**

Question 3: Check Prime Number

Buggy Code (Python)	Buggy Code (Java)
def is_prime(n): if n <= 1: return True for i in range(2, n//2): if n % i == 0: return False return True print(is_prime(7)) **# Expected Output: True**	class Main { public static boolean isPrime(int n) { if (n <= 1) return true; for (int i = 2; i < n / 2; i++) { if (n % i == 0) return false; } return true; } public static void main(String[] args) { System.out.println(isPrime(7)); **// Expected Output: true** } }

Corrected Code	Output

Question 4: Fibonacci Sequence

Buggy Code (Python)	Buggy Code (Java)
def fibonacci(n): fib = [0, 1] for i in range(2, n-1): fib.append(fib[i-1] + fib[i-2]) return fib print(fibonacci(5)) **# Expected Output: [0, 1, 1, 2, 3]**	import java.util.Arrays; class Main { public static int[] fibonacci(int n) { int[] fib = new int[n]; fib[0] = 0; fib[1] = 1; for (int i = 2; i < n-1; i++) { fib[i] = fib[i - 1] + fib[i - 2]; } return fib; } public static void main(String[] args) { System.out.println(Arrays.toString(fibonacci(5))); // **Expected Output: [0, 1, 1, 2, 3]** } }

Corrected Code	Output

Question 5: Count Vowels in a String

<table>
<tr><th>Buggy Code (Python)</th><th>Buggy Code (Java)</th></tr>
<tr><td>

```
def count_vowels(s):
    vowels = "aeiouAEIOU"
    count = 0
    for char in s:
        if vowels.contains(char):
            count += 1
        return count
print(count_vowels("hello"))

# Expected Output: 2
```

</td><td>

```
class Main
{
public static int countVowels(String s)
{
String vowels = "aeiouAEIOU";
int count = 0;
for (char ch : s.toCharArray())
{
if (vowels.contains(ch))
{
count++;
}
}
return count;
}
public static void main(String[] args)
{
System.out.println(countVowels("hello"));
// Expected Output: 2
}
}
```

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</table>

Corrected Code	Output

Question 6: Palindrome Check

Buggy Code (Python)	Buggy Code (Java)
def is_palindrome(s): return s == s.reverse() print(is_palindrome("madam")) **# Expected Output: True**	class Main { public static boolean isPalindrome(String s) { return s == new StringBuilder(s).reverse().toString(); } public static void main(String[] args) { System.out.println(isPalindrome("madam")); // **Expected Output: true** } }

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Corrected Code	Output

Question 7: Find Factorial of a Number

Buggy Code (Python)	Buggy Code (Java)
def factorial(n): if n == 1: return 1 return n * factorial(n-1) print(factorial(5)) **# Expected Output: 120**	class Main { public static int factorial(int n) { if (n == 1) return 1; return factorial(n - 1) * n; } public static void main(String[] args) { System.out.println(factorial(5)); **// Expected Output: 120** } }
Corrected Code	**Output**

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Question 8: Find Maximum in a List

<table>
<tr><th>Buggy Code (Python)</th><th>Buggy Code (Java)</th></tr>
<tr><td>

```
def find_max(arr):
        max_num = 0
        for num in arr:
            if num > max:
                max = num
        return max_num
print(find_max([3, 5, 1, 2, 4]))

# Expected Output: 5
```

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```
class Main
{
public static int findMax(int[] arr)
{
int max_num = 0;
for (int num : arr)
{
if (num > max_num)
{
max = num;
} }
return max_num;
}
public static void main(String[] args)
{
System.out.println(findMax(new int[]{3, 5, 1, 2,
4}));
// Expected Output: 5
}
}
```

</td></tr>
<tr><th>Corrected Code</th><th>Output</th></tr>
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Question 9: Check if List is Sorted

<table>
<tr><th>Buggy Code (Python)</th><th>Buggy Code (Java)</th></tr>
<tr><td>

```
def is_sorted(arr):
    for i in range(len(arr) - 1):
        if arr[i] > arr[i - 1]:
            return False
    return True
print(is_sorted([1, 2, 3, 4, 5]))
# Expected Output: True
```

</td><td>

```
class Main
{
public static boolean isSorted(int[] arr)
{
for (int i = 0; i < arr.length - 1; i++)
{
if (arr[i] > arr[i - 1])
{
return false;
}
}
return true;
}
public static void main(String[] args)
{
System.out.println(isSorted(new int[]{1, 2, 3, 4, 5}));
// Expected Output: true
}
}
```

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<tr><th>Corrected Code</th><th>Output</th></tr>
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Question 10: Count Words in a Sentence

<table>
<tr><th>Buggy Code (Python)</th><th>Buggy Code (Java)</th></tr>
<tr><td><pre>
def count_words(sentence):
    words = sentence.split()
    count = 0
    for word in words:
        count += 1
    return count + 1 # Extra addition here
print(count_words("This is a test"))
Expected Output: 4
</pre></td><td><pre>
class Main
{
public static int countWords(String sentence)
{
String[] words = sentence.split(" ");
int count = 0;
for (String word : words)
{
count += 1;
}
return count + 1; // Incorrect extra addition
}
public static void main(String[] args)
{
System.out.println(countWords("This is a test"));
// Expected Output: 4
}
}
</pre></td></tr>
<tr><th>Corrected Code</th><th>Output</th></tr>
<tr><td></td><td></td></tr>
</table>